Sample Paper 17

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

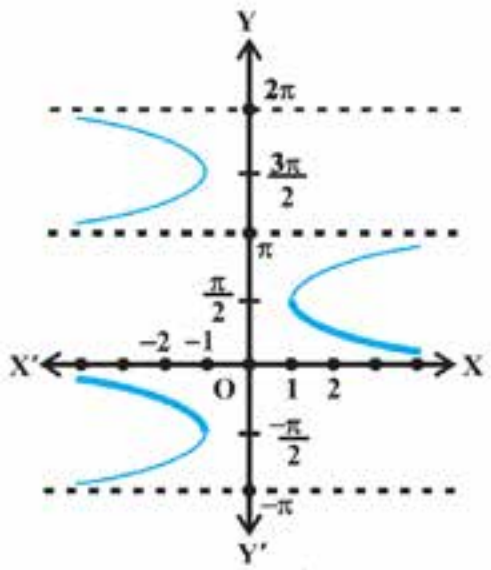
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. At which point does the graph of $y = \csc^{-1} x$ intersect the X-axis?



(a) (0,0)

(b) (1,0)

(c) (-1,0)

(d) The curve does not intersect the X-axis.

2. The figure formed by four points $\hat{i}+\hat{j}+\hat{k}$, $2\hat{i}+3\hat{j}$, $3\hat{i}-5\hat{j}-2\hat{k}$, $\hat{k}-\hat{j}$ is a

(a) Parallelogram

(b) Rectangle

(c) Trapezium

(d) Square

3. If $\vec{a}\cdot\vec{b}=0$ and $\vec{a}+\vec{b}$ makes an angle of 60° with $\vec{a}$, then

(a) $|\vec{a}|=2|\vec{b}|$

(b) $2|\vec{a}|=|\vec{b}|$

(c) $|\vec{a}|=\sqrt{3}|\vec{b}|$

(d) $\sqrt{3}|\vec{a}|=|\vec{b}|$

4. If the direction cosines of a line are $\left(\frac{1}{c},\frac{1}{c},\frac{1}{c}\right)$, then

(a) $0 < c < 1$ (b) $c > 2$

(c) $c=\pm\sqrt{2}$ (d) $c=\pm\sqrt{3}$

5. A coin and six faced die, both unbiased, are thrown simultaneously. The probability of getting a head on the coin and an odd number on the die is

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$

(c) $\frac{1}{4}$ (d) $\frac{2}{3}$

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6. Find the angle between the following pairs of lines:

$$\vec{r} = 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k})$$

and $\vec{r} = 7\hat{i} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$

(a) $\cos^{-1}\left(\frac{19}{21}\right)$ (b) $\cos^{-1}\left(\frac{16}{21}\right)$

(c) $\cos^{-1}\left(\frac{13}{21}\right)$ (d) $\cos^{-1}\left(\frac{11}{21}\right)$

7. If $P(A) = \frac{1}{12}$, $P(B) = \frac{5}{12}$ and $P\left(\frac{B}{A}\right) = \frac{1}{15}$, then $P(A \cup B)$ is equal to

(a) $\frac{89}{180}$ (b) $\frac{90}{180}$

(c) $\frac{91}{180}$ (d) $\frac{92}{180}$

8. Solution of the differential equation $xdy - ydx = 0$ represents a

(a) Parabola (b) Circle

(c) Hyperbola (d) Straight line

9. If $\int \frac{x+2}{2x^2+6x+5}dx = P\int \frac{4x+6}{2x^2+6x+5}dx + \frac{1}{2}\int \frac{dx}{2x^2+6x+5}$ Then the value of P is

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$

(c) $\frac{1}{4}$ (d) 2

10. $\int_0^{\pi/2} \left|\cos\left(\frac{x}{2}\right)\right| dx$ is equal to

(a) 1 (b) -2

(c) $\sqrt{2}$ (d) 0

11. An integrating factor of the differential equation $x\frac{dy}{dx} + y\log x = xe^x x^{-\frac{1}{2}\log x} (x > 0)$ is

(a) $x^{\log x}$ (b) $(\sqrt{x})^{\log x}$

(c) $(\sqrt{e})^{(\log x)^2}$ (d) e^{x^2}

12. For the value of λ is the function defined by

$$f(x) = \begin{cases} \lambda(x^2 - 2x), & \text{if } x \le 0 \\ 4x + 1, & \text{if } x < 0 \end{cases}$$ continuous at $x = 0$?

(a) 0

(b) 1

(c) 2

(d) Not continuous at $x = 0$ for any value of λ

13. If gas is being pumped into a spherical balloon at the rate of 30 ft3/min. Then, the rate at which the radius increases, when it reaches the value 15 ft is

(a) $\frac{1}{15\pi}$ ft/min (b) $\frac{1}{30\pi}$ ft/min

(c) $\frac{1}{20}$ ft/min (d) $\frac{1}{25}$ ft/min

14. The length of the longest interval, in which $f(x) = 3\sin x - 4\sin^3 x$ is increasing is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

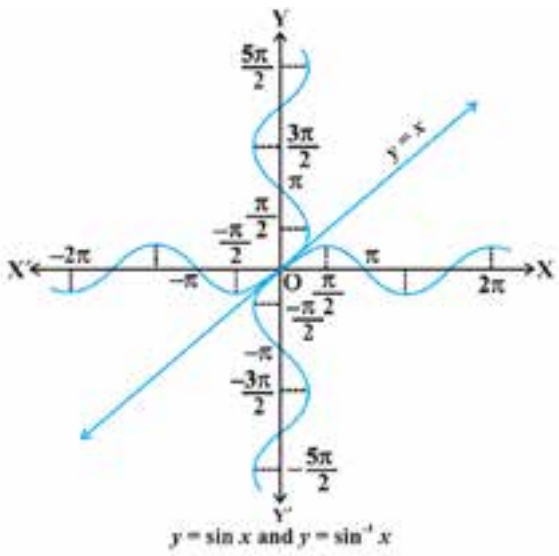
(c) $\frac{3\pi}{2}$ (d) π

15. If $A(\alpha) = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$, then the matrix $A^2(\alpha)$ is equal to

(a) $A(2\alpha)$ (b) $A(\alpha)$

(c) $A(3\alpha)$ (d) $A(4\alpha)$

16. What happens to the value of $y = \tan x$ as x approaches $\pi/2$ from the left side?

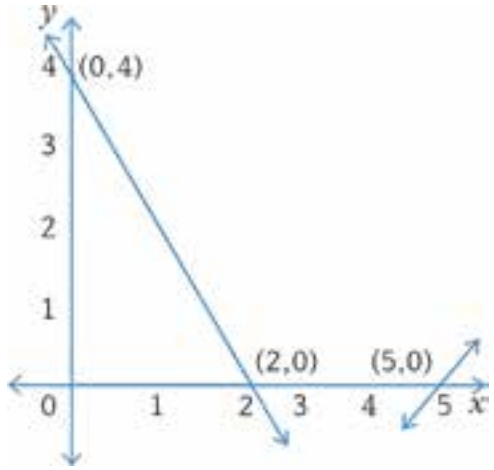


(a) $y \to 0$ (b) $y \to \infty$

(c) $y + \infty$ (d) Undefined

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17. For the same LPP with objective function $Z = 3x + 2y$, what is the minimum value of Z and at which point does it occur?



(a) Z=8 at (0,4) (b) Z=15 at (5,0)

(c) Z=6 at (2,0) (d) No minimum exists

18. Solution of $e^{dy/dx} = x$, when $x = 1$ and $y = 0$ is

(a) $y = x(\log x - 1) + 4$

(b) $y = x(\log x - 1) + 3$

(b) $y = x(\log x + 1) + 1$

(d) $y = x(\log x - 1) + 1$

19. If $n > 1$, then

Assertion : $\int_0^{\infty} \frac{dx}{1+x^n} = \int_0^{1} \frac{dx}{(1-x^n)^{1/n}}$

Reason : $\int_a^b f(x)\,dx = \int_a^b f(a+b+x)\,dx$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

20. **Assertion :** $\int_{-\pi/2}^{\pi/2} |\sin x|\,dx = 2$

Reason : $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, where c ε a, b)

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. If a line makes angles 90°, 135°, 45° with then x, y and z axes respectively, find the direction consines.

22. If $P(not\ A) = 0.7$, $P(B) = 0.7$ and $P(B/A) = 0.5$, then find $P(A/B)$.

23. Find $\int \frac{\sin^2 x - \cos^2 x}{\sin x \cos x}\,dx$.

OR

Find $\int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x}\,dx$.

24. Find the position vector of a point which divides the join of points with position vectors $\vec{a} - 2\vec{b}$ and $2\vec{a} + \vec{b}$ externally in the ratio 2 : 1.

OR

If $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$, then find a unit vector parallel to the vector $\vec{a} + \vec{b}$.

25. If $R = \{(a, a^3)$: a is a prime number less than 5$\}$ be a relation. Find the range of R.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. The total cost $C(x)$ associated with the production of x units of an item is given by $C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$.

Find the marginal cost when 3 units are produced, where by marginal cost we mean the instantaneous rate of change of total cost at any level of output.

OR

The total revenue received from the sale of x units of a product is given by $R(x) = 3x^2 + 36x + 5$ in rupees. Find the marginal revenue we mean the rate of change of total revenue with respect to the number of times sold at an instant.

27. Evaluate $\int \frac{\cos\sqrt{x}}{\sqrt{x}}\,dx$.

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28. If $\hat{a}$, $\hat{b}$ and $\hat{c}$ are mutually perpendicular unit vectors, then find the value of $|2\hat{a}+\hat{b}+\hat{c}|$.

29. Determine the value of k for which the following function is continuous at $x=3$:

$$f(x)=\begin{cases}\dfrac{(x+3)^2-36}{x-3}, & x\neq 3\\ k, & x=3\end{cases}$$

OR

Determine the value of constant k so that the function $f(x)=\begin{cases}\dfrac{kx}{|x|}, & \text{if } x<0\\ 3, & \text{if } x\geq 0\end{cases}$ is continuous at $x=0$.

30. Find $|AB|$, if $A=\begin{bmatrix}0 & -1\\ 0 & 2\end{bmatrix}$ and $B=\begin{bmatrix}3 & 5\\ 0 & 0\end{bmatrix}$.

31. Write the value of $\tan^{-1}\left[2\sin\left(2\cos^{-1}\dfrac{\sqrt{3}}{2}\right)\right]$.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Find the particular solution of the differential equation $\dfrac{dy}{dx}=\dfrac{xy}{x^2+y^2}$ given that $y=1$, when $x=0$.

33. Maximize $Z=3x+4y$, subject to the constraints; $x+y\leq 4$, $x\geq 0$, $y\geq 0$.

OR

Minimize $Z=-3x+4y$ subject to the constraints

$$x+2y \leq 8,$$
$$3x+2y \leq 12,$$
$$x \geq 0,\ y\geq 0$$

34. Find : $\int \dfrac{3x+5}{x^2+3x-18}\,dx$.

OR

Prove that $\int_0^a f(x)\,dx=\int_0^a f(a-x)\,dx$, hence evaluate $\int_0^{\pi}\dfrac{x\sin x}{1+\cos^2 x}\,dx$.

35. Prove that the line through $A(0,-1,-1)$ and $B(4,5,1)$ intersects the line through $C(3,9,4)$ and $D(-4,4,4)$.

SECTION - E

Case study based questions are compulsory.

36. OYO Rooms, also known as OYO Hotels & Homes, is an Indian multinational hospitality chain of leased and franchised hotels, homes and living spaces. Founded in 2012 by Ritesh Agarwal, OYO initially consisted mainly of budget hotels.



Data analyst at OYO say that during frequent trips to a certain city, a traveling salesperson stays at hotel A 50% of the time, at hotel B 30% of the time, and at hotel C 20% of the time. When checking in, there is some problem with the reservation 3% of the time at hotel A, 6% of the time at hotel B, and 10% of the time at hotel C. Suppose the salesperson travels to this city.

(i) Find the probability that the salesperson stays at hotel A and has a problem with the reservation.

(ii) Find the probability that the salesperson has a problem with the reservation.

(iii) Suppose the salesperson has a problem with the reservation; what is the probability that the salesperson is staying at hotel A?

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37. A car carrier trailer, also known as a car-carrying trailer, car hauler, or auto transport trailer, is a type of trailer or semi-trailer designed to efficiently transport passenger vehicles via truck. Commercial-size car carrying trailers are commonly used to ship new cars from the manufacturer to auto dealerships. Modern car carrier trailers can be open or enclosed. Most commercial trailers have built-in ramps for loading and off-loading cars, as well as power hydraulics to raise and lower ramps for stand-alone accessibility.



A transport company uses three types of trucks T_1, T_2 and T_3 to transport three types of vehicles V_1, V_2 and V_3. The capacity of each truck in terms of three types of vehicles is given below:

	V_1	V_2	V_3
T_1	1	3	2
T_2	2	2	3
T_3	3	2	2

Using matrix method find:

(i) The number of trucks of each type required to transport 85, 105 and 110 vehicles of V_1, V_2 and V_3 types respectively.

(ii) Find the number of vehicles of each type which can be transported if company has 10, 20 and 30 trucks of each type respectively.

38. Hindustan Pencils Pvt. Ltd. is an Indian manufacturer of pencils, writing materials and other stationery items, established in 1958 in Bombay. The company makes writing implements under the brands Nataraj and Apsara, and claims to be the largest pencil manufacturer in India.

HINDUSTAN PENCILS PVT. LTD. apsara NATARAJ

Hindustan Pencils manufactures x units of pencil in a given time, if the cost of raw material is square of the pencils produced, cost of transportation is twice the number of pencils produced and the property tax costs ₹ 5000. Then,

(i) Find the cost function $C(x)$.

(ii) Find the cost of producing 21st pencil.

(iii) The marginal cost of producing 50 pencils.

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